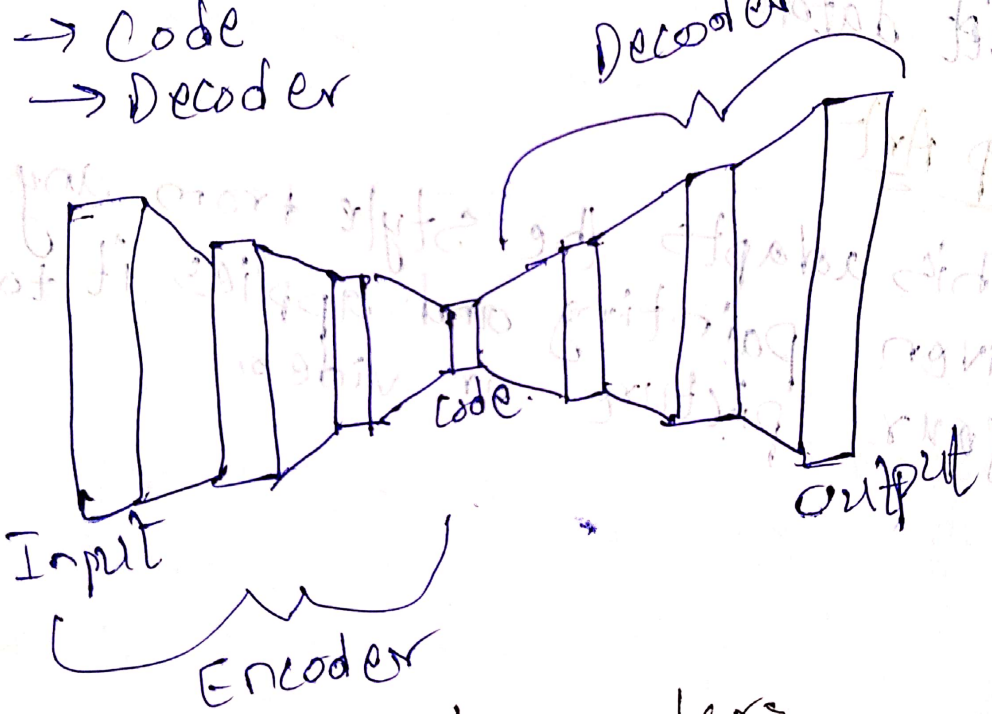
CO-4

Autoencoders

* Autoencoders are a type of deep learning algorithm that is designed to receive an input and transform it into a different representation.

Three layers
→ Encoder
→ Code
→ Decoder

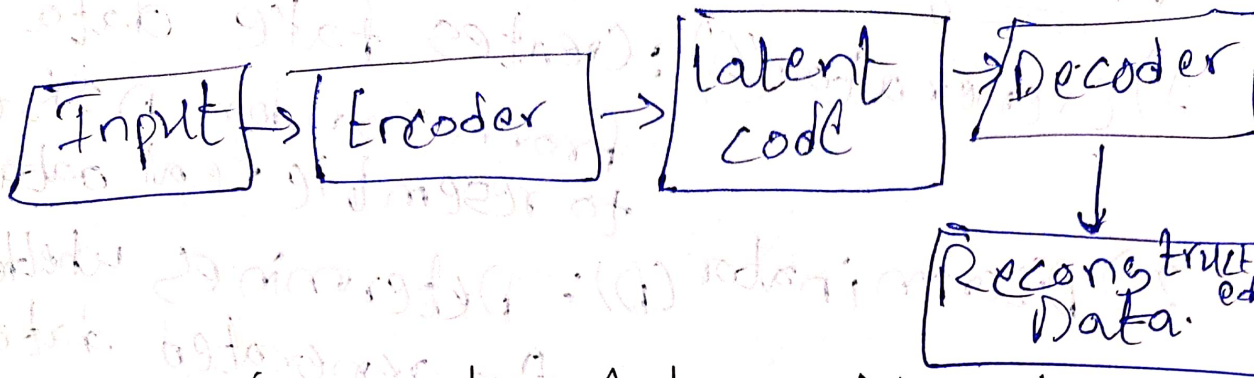
* Denoising Autoencoder
* Sparse Autoencoder
* Convolutional
* Variational



Variational Autoencoders

* A VAE comprises an encoder network that maps input data to a latent code & decoder network that conducts the inverse operation by translating the latent code back to the

reconstruction data.



GAN (Generative Adversarial Networks)

* These are algorithmic architectures that use two neural networks, pitting one against the other in order to generate new, synthetic instances of data.

Use Cases

→ Change facial expressions.
→ create computer game scenes.
→ Visualize designs.

Generative Models

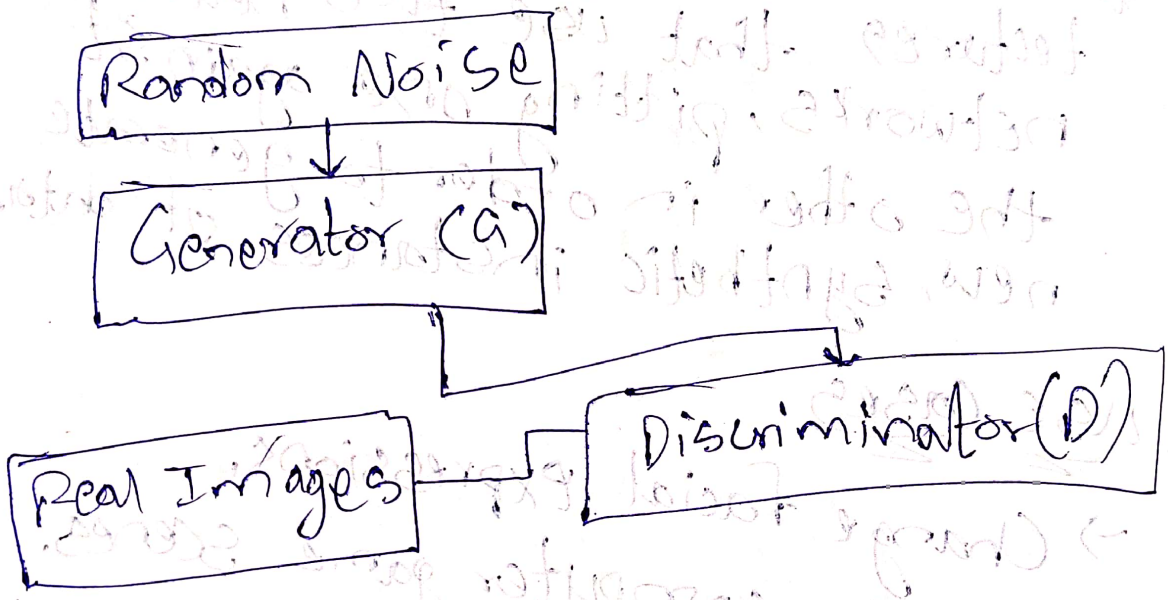
* It explicitly models the actual distribution of the class.
* It learns the joint probability function. $p(x|y)$.

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Working

1) Generator (G): Creates fake data from random noise to resemble real data.

2) Discriminator (D): Determines whether the generated data is real or fake



Natural Language Processing (NLP)

* Branch of AI that enables computers to understand, interpret, analyze, generate & respond to human language in both text & speech forms.

Components

1) Text Preprocessing.

2) Tokenization.

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3) Stop Word Removal
4) Stemming
5) Lemmatization.
6) Feature Extraction
7) Natural Language Understanding (NLU)
8) Natural Language Generation (NLG)

Applications

1) Chatbots
2) Machine Translation
3) Sentiment Analysis.
4) Spam Detection
5) Voice Assistants.

Speech Analysis

*Process of examining and interpreting human speech using signal processing, AI & machine learning techniques.

Applications

→ Virtual Voice Assistants
→ Healthcare
→ Banking Authentication

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GAN in REAL Time Applications

→ Image Generation & Editing.
→ Medical Imaging.
→ Image Super-Resolution.
→ Autonomous Vehicles.
→ Speech & voice Generation.
→ Gaming & virtual reality.
→ Data Augmentation.
→ Cybersecurity.

Recommendation System

* This is an intelligent info Filtering system that predict the preferences or interests of users & suggests relevant items such as products, movies, books, music, restaurants, news or online courses.

User Data Collection
↓
Data Preprocessing
↓
Feature Extraction
↓
Recommendation Algorithm

↓

Ranking of items.

↓

Top - N Recommendations

↓

User Feed back.

Types

1) Content - Based filtering.

2) Collaborative Filtering.

3) Hybrid

4) Knowledge - Based

5) Demographic => Age, Gender.

6) Context - Aware => Time, Weather, Location.

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